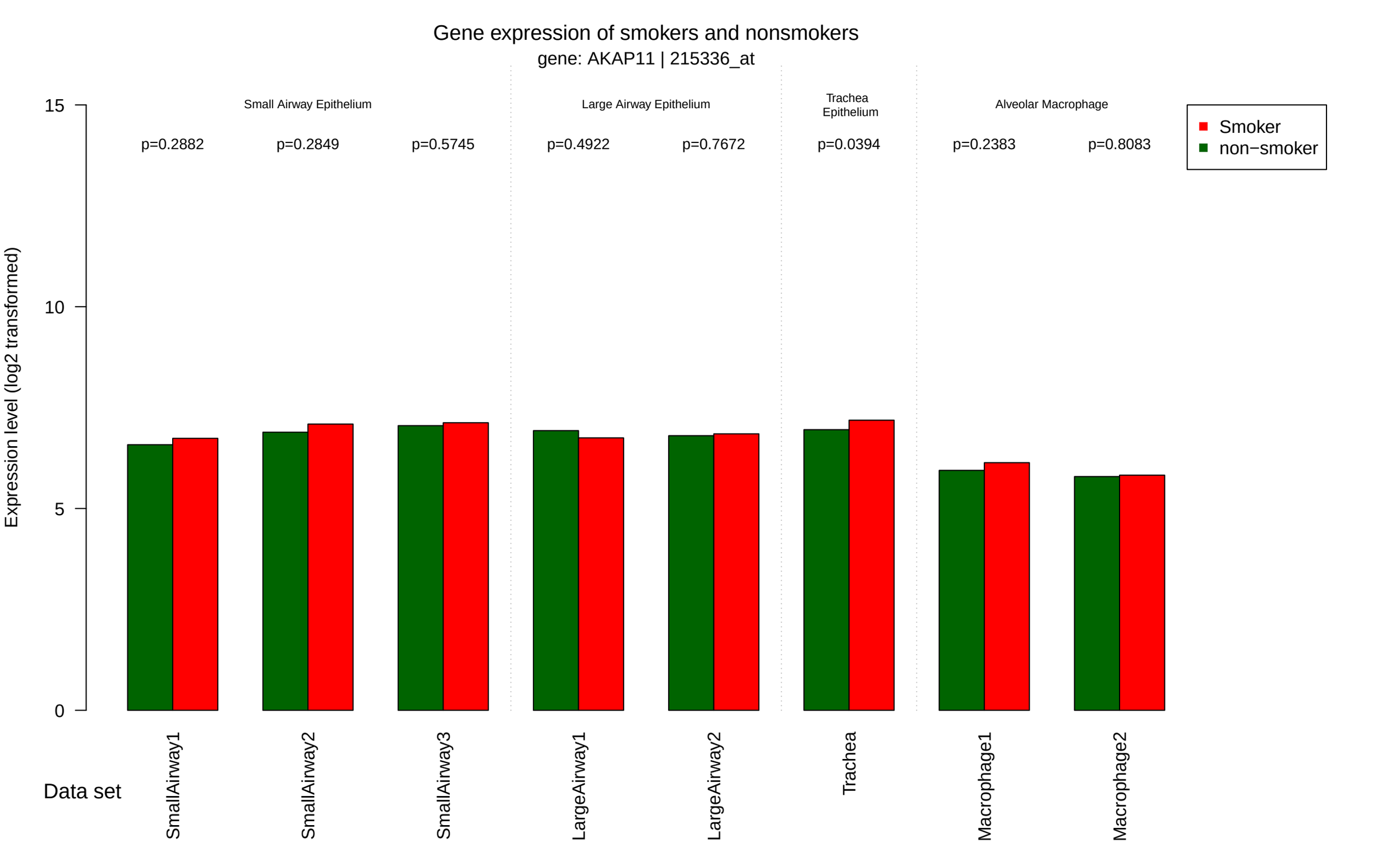
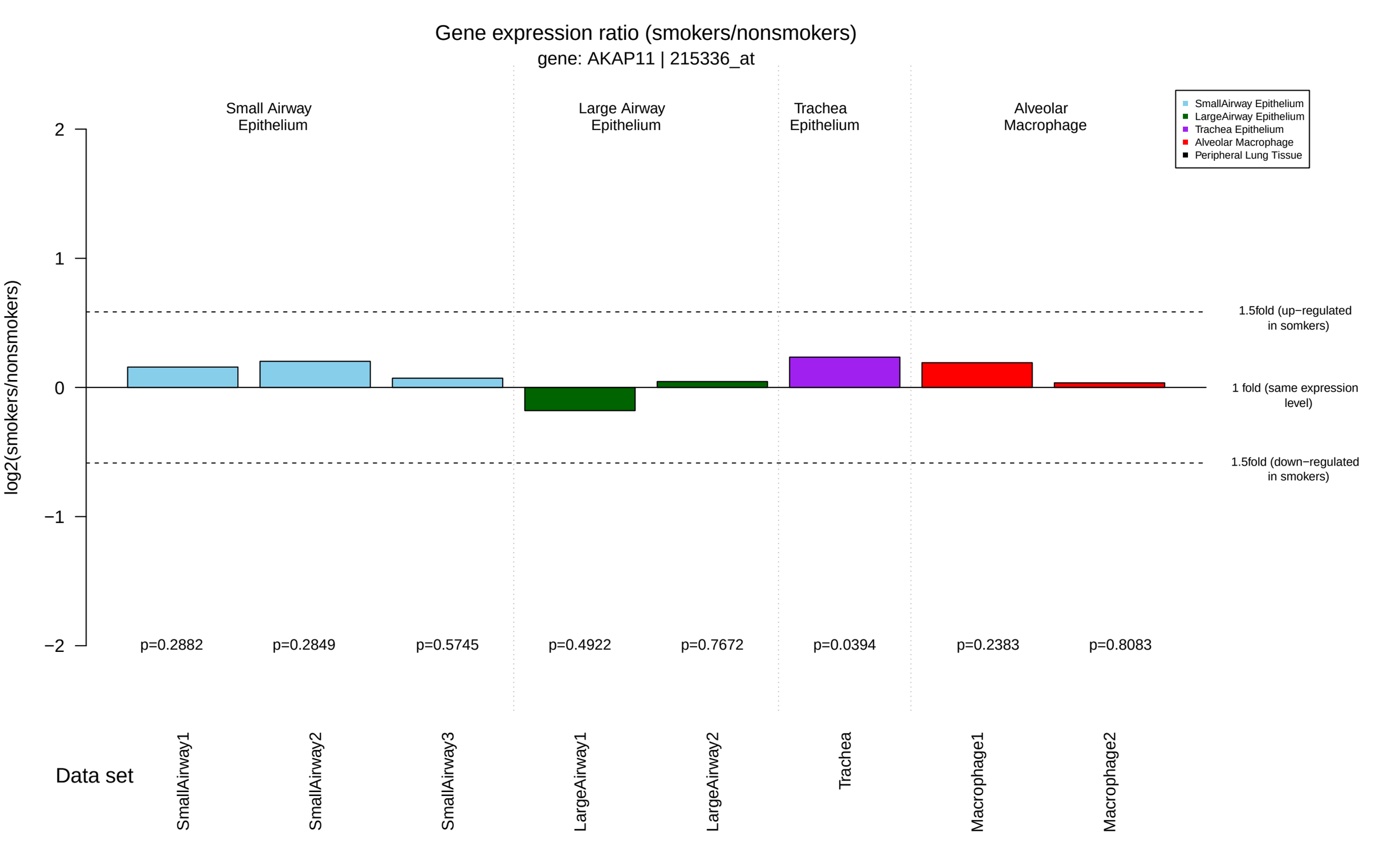
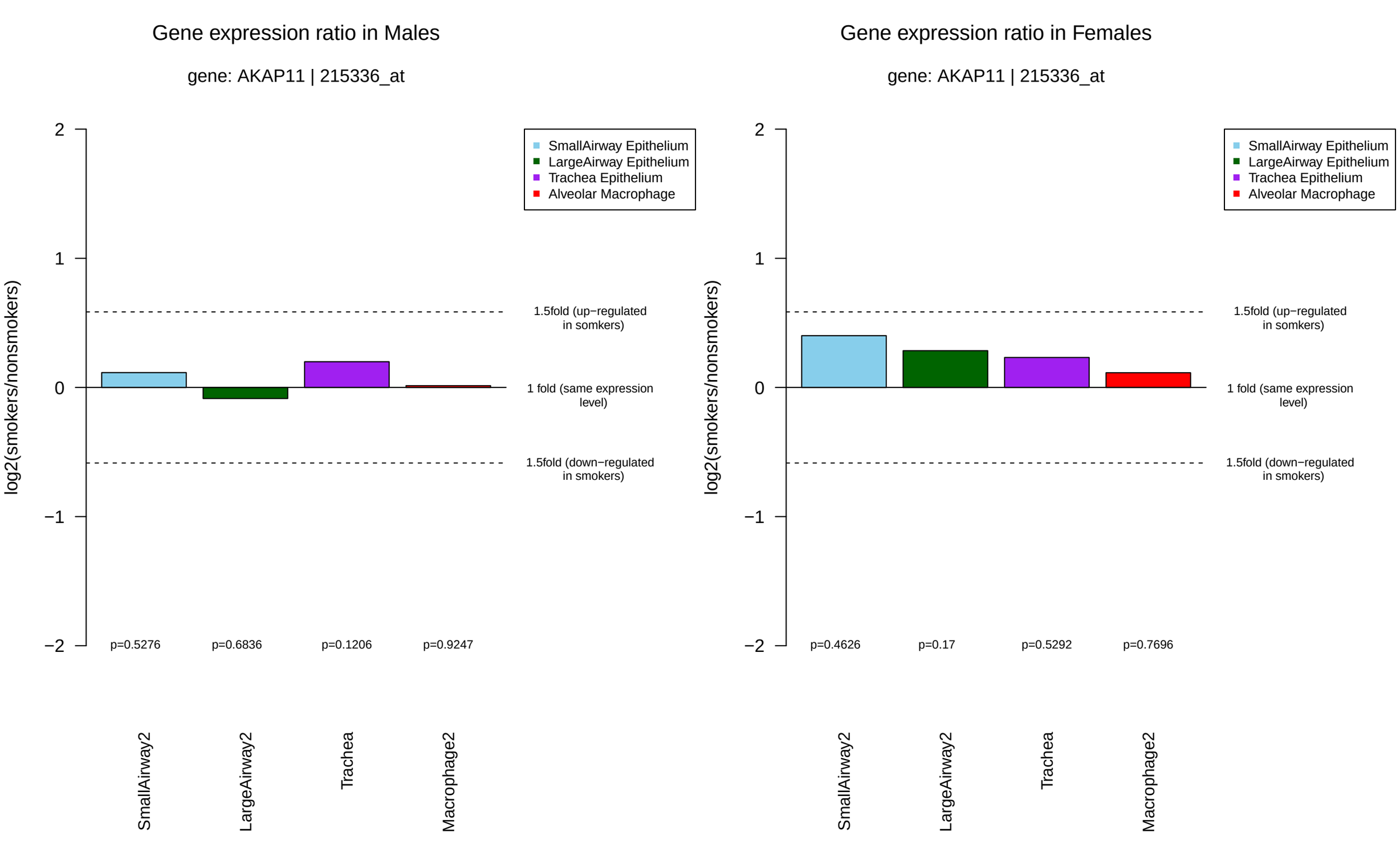
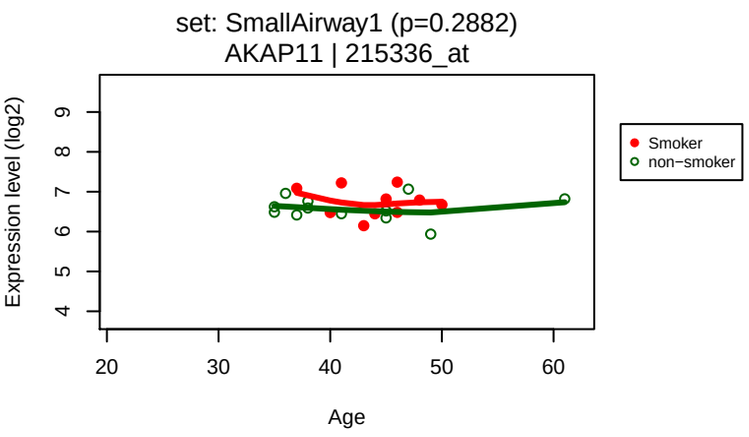
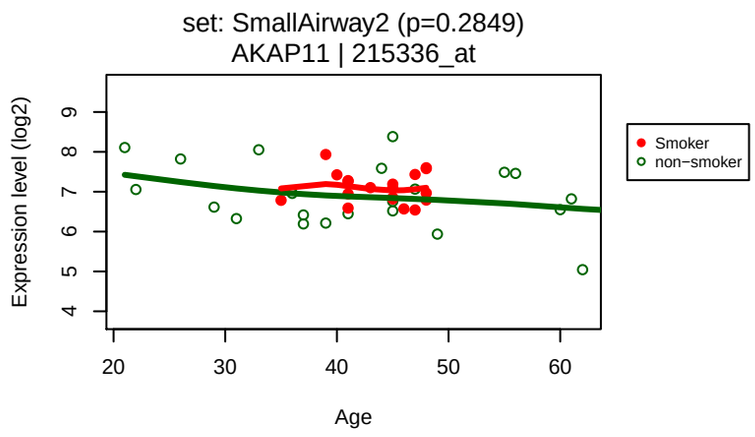
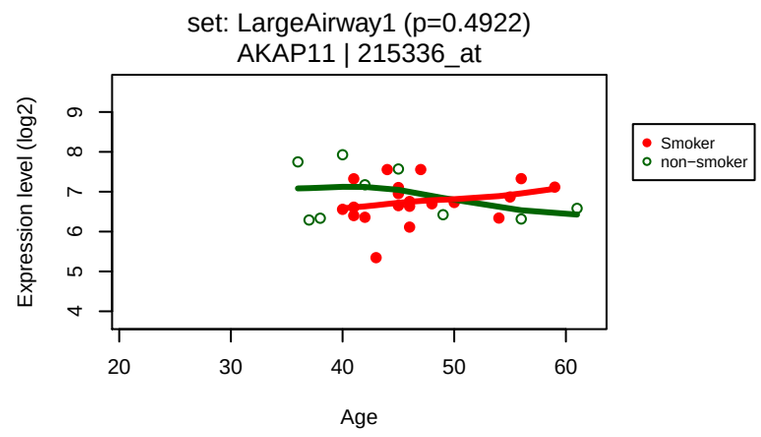

Gene expression of smokers and nonsmokers
gene: AKAP11 | 215336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2882
p=0.2849
p=0.5745
p=0.4922
p=0.7672
p=0.0394
p=0.2383
p=0.8083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

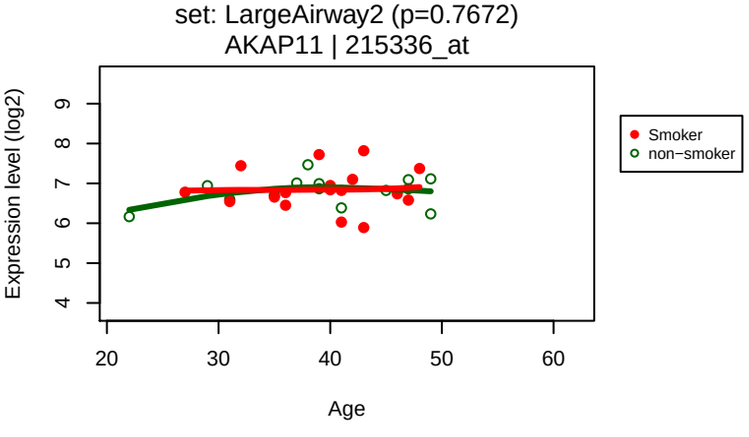

Gene expression ratio (smokers/nonsmokers)
gene: AKAP11 | 215336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2882
p=0.2849
p=0.5745
p=0.4922
p=0.7672
p=0.0394
p=0.2383
p=0.8083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

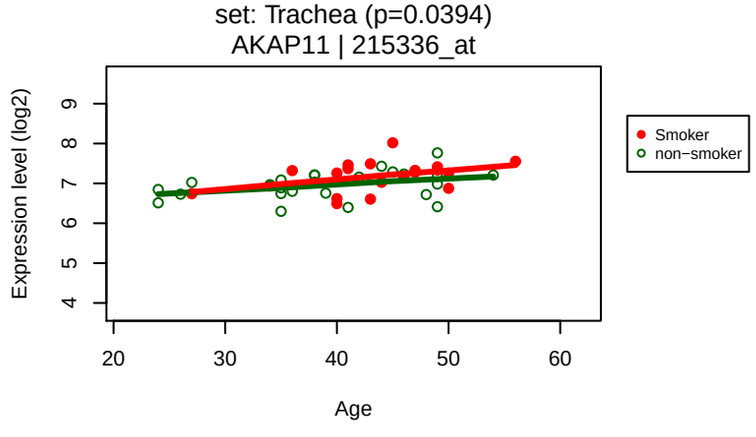

Gene expression ratio in Males
gene: AKAP11 | 215336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5276
p=0.6836
p=0.1206
p=0.9247
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP11 | 215336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4626
p=0.17
p=0.5292
p=0.7696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

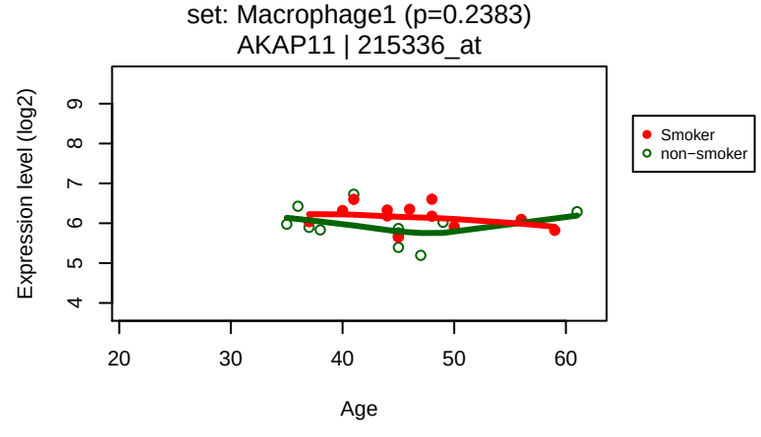

set: SmallAirway1 (p=0.2882)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker

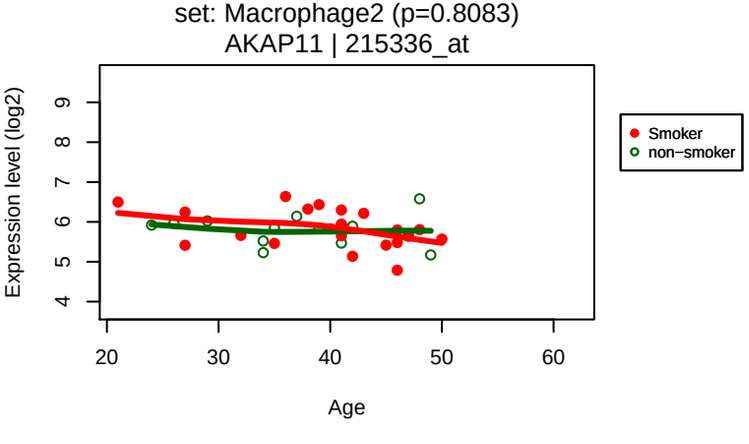

set: SmallAirway2 (p=0.2849)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker

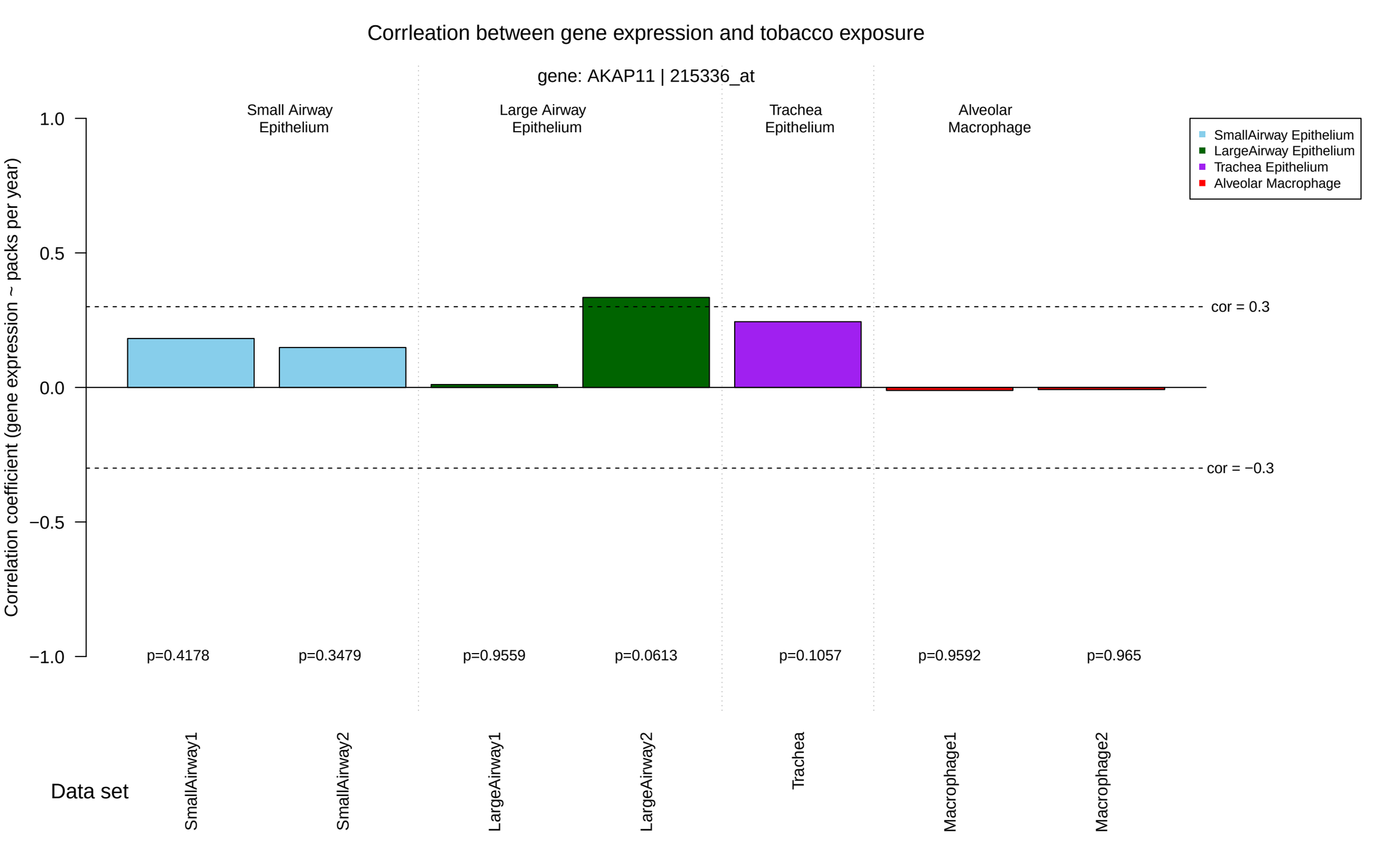

set: LargeAirway1 (p=0.4922)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7672)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0394)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2383)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8083)
AKAP11 | 215336_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP11 | 215336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4178
p=0.3479
p=0.9559
p=0.0613
p=0.1057
p=0.9592
p=0.965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2